

# A Fixed Covering Certificate for Subset Sums of Powers of 3, 4, 7

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## Abstract

Let

$$B = \{3^a : a \geq 1\} \cup \{4^b : b \geq 1\} \cup \{7^c : c \geq 1\}.$$

We prove that every integer  $n \geq 582$  is representable as a subset sum of distinct elements of  $B$ . The proof combines a finite subset-sum interval certificate with an interval-overlap propagation lemma. The principal finite check is a weighted gap lemma for later powers of 3, 4, 7, verified by a replayable modular certificate. The result is a fixed-case covering theorem only; no optimality of the endpoint is claimed.

## 1 Introduction

Questions about which integers are representable as subset sums of a prescribed sequence are a classical elementary theme. For nondecreasing sequences beginning with 1, Brown's criterion gives a well-known interval-overlap condition for completeness; see [1]. The present note treats a narrower eventual covering problem for a sparse sequence that does not begin with 1.

Let

$$B = \{3^a : a \geq 1\} \cup \{4^b : b \geq 1\} \cup \{7^c : c \geq 1\}.$$

We prove that every integer at least 582 is a subset sum of distinct elements of  $B$ . The point of recording the result is not the size of the endpoint, and no minimality assertion is made. Rather, the proof gives a compact fixed-case certificate: a finite interval of coverage is verified directly, and a finite modular certificate verifies that the overlap condition continues for every later power in the sequence.

The proof begins with a finite interval certificate

$$[582, 4460] \subset \text{SubSum}(\{3, 4, 7, 9, 16, 27, 49, 64, 81, 243, 256, 343, 729, 1024, 2187\}),$$

then uses an interval-overlap lemma to propagate coverage indefinitely. The propagation step is reduced to a weighted gap lemma for the next powers of 3, 4, 7. That lemma is verified by a finite modular certificate supplied with the paper.

## 2 Definitions

Let  $B$  be ordered increasingly as

$$b_1 < b_2 < b_3 < \cdots.$$

For a finite set  $S$  of integers, define

$$\text{SubSum}(S) = \left\{ \sum_{x \in T} x : T \subseteq S \right\}.$$

We write

$$A_1(\{3, 4, 7\}) = \text{SubSum}(B).$$

For  $t \in B$ , define

$$S(t) = \sum_{\substack{p \in B \\ p < t}} p.$$

For  $t > 2187$ , define the next-power gaps

$$\Delta_3(t) = \min\{3^a : 3^a > t\} - t,$$

$$\Delta_4(t) = \min\{4^b : 4^b > t\} - t,$$

and

$$\Delta_7(t) = \min\{7^c : 7^c > t\} - t.$$

### 3 The fixed covering theorem

**Theorem 1.** *Let*

$$B = \{3^a : a \geq 1\} \cup \{4^b : b \geq 1\} \cup \{7^c : c \geq 1\}.$$

*Then*

$$[582, \infty) \subset A_1(\{3, 4, 7\}).$$

*Equivalently, every integer  $n \geq 582$  is representable as a subset sum of distinct elements of  $B$ .*

### 4 The initial interval certificate

The first fifteen elements of  $B$  are

$$3, 4, 7, 9, 16, 27, 49, 64, 81, 243, 256, 343, 729, 1024, 2187.$$

A finite subset-sum enumeration verifies

$$[582, 4460] \subset \text{SubSum}(\{3, 4, 7, 9, 16, 27, 49, 64, 81, 243, 256, 343, 729, 1024, 2187\}).$$

Set

$$M = 582, \quad N = 4460.$$

The sum of the first fifteen listed terms is 5042, so  $N = 5042 - M$ .

### 5 Interval-overlap propagation

**Lemma 1** (Interval-overlap propagation). *Let*

$$[M, N] \subset \text{SubSum}(S)$$

*for a finite set  $S$  of distinct allowed terms. Let  $p \notin S$  be a new allowed term. If*

$$p \leq N - M + 1,$$

*then*

$$[M, N + p] \subset \text{SubSum}(S \cup \{p\}).$$

*Proof.* The interval  $[M, N]$  is already covered by subset sums from  $S$ . Adding  $p$  to each representation covers the interval

$$[M + p, N + p].$$

The condition  $p \leq N - M + 1$  is equivalent to

$$M + p \leq N + 1.$$

Hence  $[M, N]$  and  $[M + p, N + p]$  overlap or touch. Their union therefore covers  $[M, N + p]$ , proving the claim.  $\square$

Suppose the current covered interval has the form  $[M, N]$ , where  $M = 582$ , and the next unused power is  $t \in B$ . In the propagation below,  $N = S(t) - M$ , so the condition  $t \leq N - M + 1$  is equivalent to

$$S(t) - t \geq 2M - 1 = 1163.$$

It remains to prove this inequality for every  $t > 2187$  in  $B$ .

## 6 Weighted next-gap identities

**Lemma 2** (Weighted next-gap identities). *For every future power  $t > 2187$  generated by 3, 4, 7, the following identities hold.*

*If  $t = 3^r$ ,  $r \geq 8$ , then*

$$S(t) - t - 1163 = \frac{2\Delta_4(t) + \Delta_7(t) - 7002}{6}.$$

*If  $t = 4^s$ ,  $s \geq 6$ , then*

$$S(t) - t - 1163 = \frac{3\Delta_3(t) + \Delta_7(t) - 7002}{6}.$$

*If  $t = 7^q$ ,  $q \geq 4$ , then*

$$S(t) - t - 1163 = \frac{3\Delta_3(t) + 2\Delta_4(t) - 7002}{6}.$$

*Proof.* We prove the first identity; the other two follow by the same calculation. Let  $t = 3^r$ , and let

$$4^a = 3^r + \Delta_4(t), \quad 7^b = 3^r + \Delta_7(t)$$

be the least larger powers of 4 and 7, respectively. Since the powers below  $3^r$  are finite geometric sums on the three bases,

$$S(3^r) = \frac{3^r - 3}{2} + \frac{4^a - 4}{3} + \frac{7^b - 7}{6}.$$

Substituting  $4^a = 3^r + \Delta_4(t)$  and  $7^b = 3^r + \Delta_7(t)$  gives

$$S(3^r) - 3^r - 1163 = \frac{2\Delta_4(t) + \Delta_7(t) - 7002}{6}.$$

The identities for  $t = 4^s$  and  $t = 7^q$  are obtained similarly, using the least larger powers on the two competing bases.  $\square$

## 7 The weighted gap lemma

**Lemma 3** (Weighted gap lemma). *For every future power  $t > 2187$  generated by 3, 4, 7, the corresponding weighted gap inequality holds:*

$$t = 3^r, \ r \geq 8 \implies 2\Delta_4(t) + \Delta_7(t) \geq 7002,$$

$$t = 4^s, \ s \geq 6 \implies 3\Delta_3(t) + \Delta_7(t) \geq 7002,$$

and

$$t = 7^q, \ q \geq 4 \implies 3\Delta_3(t) + 2\Delta_4(t) \geq 7002.$$

*Proof.* The proof is by a finite modular certificate. A failure of the first inequality would give a bounded full paired system

$$4^a - 3^r = u, \quad 7^b - 3^r = v, \quad 2u + v \leq 7001, \quad r \geq 8,$$

with  $1 \leq u \leq 3500$  and  $1 \leq v \leq 7000$ .

A failure of the second inequality would give

$$3^a - 4^s = u, \quad 7^b - 4^s = v, \quad 3u + v \leq 7001, \quad s \geq 6,$$

with  $1 \leq u \leq 2333$  and  $1 \leq v \leq 7000$ .

A failure of the third inequality would give

$$3^a - 7^q = u, \quad 4^b - 7^q = v, \quad 3u + 2v \leq 7001, \quad q \geq 4,$$

with  $1 \leq u \leq 2333$  and  $1 \leq v \leq 3500$ .

The residue filters are finite. In the first case,  $r \geq 8$  implies

$$3^r \equiv 0 \pmod{6561},$$

so every failure must satisfy

$$4^a \equiv u \pmod{6561}, \quad 7^b \equiv v \pmod{6561}.$$

In the second case,  $s \geq 6$  implies

$$4^s \equiv 0 \pmod{4096},$$

so every failure must satisfy

$$3^a \equiv u \pmod{4096}, \quad 7^b \equiv v \pmod{4096}.$$

In the third case,  $q \geq 4$  implies

$$7^q \equiv 0 \pmod{2401},$$

so every failure must satisfy

$$3^a \equiv u \pmod{2401}, \quad 4^b \equiv v \pmod{2401}.$$

The accompanying replayable certificate constructs all residue-admissible systems and eliminates them by full paired modular tests. The initial residue-admissible counts are

$$1, 361, 889, \quad 255, 938, \quad 1, 501, 666$$

in the three cases, respectively. In each case, the final number of surviving systems is zero. The certificate checks the full paired systems displayed above, not merely the resulting difference equations. Therefore every possible failure system is excluded, and the weighted gap inequalities hold.  $\square$

## 8 Proof of the theorem

The finite interval certificate gives

$$[582, 4460] \subset \text{SubSum}(b_1, \dots, b_{15}),$$

where

$$b_1, \dots, b_{15} = 3, 4, 7, 9, 16, 27, 49, 64, 81, 243, 256, 343, 729, 1024, 2187.$$

Let  $t = b_j > 2187$  be any later power in the ordered sequence  $B$ . By the weighted gap lemma and the weighted next-gap identities,

$$S(t) - t \geq 1163.$$

As observed above, this is exactly the interval-overlap condition for adjoining the next power  $t$ . Hence the interval-overlap lemma applies successively to every later power. Starting from  $[582, 4460]$ , the covered interval propagates without bound. Therefore

$$[582, \infty) \subset A_1(\{3, 4, 7\}).$$

This proves the theorem.

## 9 Computational certificate

The supplementary files contain a replayable computational certificate. The certificate checks two finite claims.

First, it checks the initial subset-sum interval certificate

$$[582, 4460] \subset \text{SubSum}(\{3, 4, 7, 9, 16, 27, 49, 64, 81, 243, 256, 343, 729, 1024, 2187\}).$$

This is finite because only  $2^{15}$  subset sums are involved.

Second, it checks the weighted gap lemma. Each possible failure of the lemma gives one of the bounded full paired systems described in the proof. For example, in Case A a failure is represented by integers  $u, v$  satisfying  $1 \leq u \leq 3500$ ,  $1 \leq v \leq 7000$ , and  $2u + v \leq 7001$ , together with the full paired system

$$4^a - 3^r = u, \quad 7^b - 3^r = v, \quad r \geq 8.$$

The other two cases are analogous. Thus each case begins with finitely many bounded pairs  $(u, v)$ .

For each bounded pair, the verifier first applies the stabilized residue filter modulo 6561, 4096, or 2401, depending on the case. This produces a finite list of residue-admissible systems, with forced exponent classes modulo the corresponding multiplicative orders. The verifier then applies the moduli listed in the certificate tables. A row is eliminated only when the relevant congruences for the full paired system have no simultaneous solution modulo the selected modulus.

This last point is important: the verifier does not merely test the difference equation obtained by subtracting the two displayed equations. It checks the paired congruence system involving the common baseline power. The certificate therefore excludes the full possible failure systems required by the weighted gap lemma.

The certificate status is:

locally replayed; replayable computational certificate included.

No independent third-party acceptance of the certificate is asserted here.

To replay the certificates, run:

```
python3 verify_initial_interval.py
python3 verify_weighted_gap_certificates.py
```

The expected weighted-gap terminal summary is:

```
WEIGHTED GAP CERTIFICATE REPLAY | PASS
Case A: initial = 1,361,889; final survivors = 0
Case B: initial = 255,938; final survivors = 0
Case C: initial = 1,501,666; final survivors = 0
Total residue-admissible systems checked = 3,119,493
Weighted Gap Lemma: PASS
Global overlap inequality: PASS
Fixed theorem chain: PASS
FINAL STATUS: PASS | WEIGHTED_GAP_PROVED
```

## 10 Limitations

This paper proves only the fixed case  $D = \{3, 4, 7\}$  with  $k = 1$ . It does not claim a result for any other generating set. It does not claim a result for  $k \neq 1$ . It does not claim that 582 is the least possible endpoint. It does not claim a density theorem, an asymptotic theorem, or a classification theorem.

## References

- [1] J. L. Brown, Jr., *Notes on complete sequences of integers*, American Mathematical Monthly **68** (1961), 557–560.